Prefix: Substituents that are attached to the parent carbon chain are denoted by adding prefix names before the root word. The prefix names for some common substituents are listed below. If the functional groups are not part of the parent chain, they are considered as substituents. In such cases its prefix name is added before the root word. Prefix names for some functional groups mentioned along with their secondary prefix are listed in table 11.4

Table 11.5 List of substitutents and their Prefix names

Substituent group	Prefix
-F	Fluoro
-Cl	Chloro
-Br	Bromo
-I	Iodo
$-NO_2$	Nitro
-NO	Nitroso
–N=N–	Diazo
-OR	Alkoxy
$-OCH_3$ (or) -OMe	Methoxy
$-OC_2H_5$ (or) -OEt	Ethoxy
$-CH_3$ (or) –Me	Methyl
$-C_2H_5$ (or) -Et	Ethyl
$-CH_2-CH_2-CH_3$	Propyl
$-CH(CH_3)_2$	Propan-2-yl (isopropyl)
$-CH_2-CH_2-CH_2-CH_3$	Butyl
$-CH_2-CH(CH_3)-CH_3$	2-methyl propyl (isobutyl)
$-C(CH_3)_3$	2-methylpropan-2-yl (tertbutyl)
$-CH_2-CH_2-CH_2-CH_2-CH_3$	Pentyl
$-CH_2-CH(CH_3)-CH_2-CH_3$	2-methyl butyl
$-CH_2-C(CH_3)_3$	2,2-dimethyl propyl (neopentyl)

11.3.1 IUPAC rules for nomenclature of organic compounds

The following steps should be followed for naming an organic compound as per IUPAC nomenclature.

1. Choose the longest carbon chain. (Root word). Consider all the other groups attached to this chain as substitutents.

2. Numbering of the longest carbon chain

3. Naming of the substituents (prefixes or suffixes)

4. Arrange the substitutents in the alphabetical order

5. Write the name of the compound as below

"prefix + root word + primary suffix + secondary suffix"

Unit 11 revised - jagan.indd 116 22-01-2021 12:42:17